

## 01.05 The Normal Distribution

### Practice problems

Show your work and state restrictions or excluded values when needed.

| Recognize                                                    | Apply these rules                                                                                              |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Statistics And Probability                                   | Identify the population, variable, or probability model before calculating.                                    |
| The expression contains the chapter's key notation or model. | Write the known quantities, choose the matching formula, calculate carefully, and check the result in context. |

### A. Practice: The Normal Distribution

1.

$$P(A) = \frac{3}{10}$$

2.

$$\mu = \frac{12 + 15 + 18 + 20}{4}$$

3.

$$Z = \frac{82 - 70}{8}$$

4.

$$P(A) = \frac{3}{10}$$

5.

$$\mu = \frac{12 + 15 + 18 + 20}{4}$$

6.

$$Z = \frac{82 - 70}{8}$$

7.

$$P(A) = \frac{3}{10}$$

8.

$$\mu = \frac{12 + 15 + 18 + 20}{4}$$

9.

$$Z = \frac{82 - 70}{8}$$

10.

$$P(A) = \frac{3}{10}$$

11.

$$\mu = \frac{12 + 15 + 18 + 20}{4}$$

12.

$$Z = \frac{82 - 70}{8}$$

13.

$$P(A) = \frac{3}{10}$$

14.

$$\mu = \frac{12 + 15 + 18 + 20}{4}$$

15.

$$Z = \frac{82 - 70}{8}$$

16.

$$P(A) = \frac{3}{10}$$

17.

$$\mu = \frac{12 + 15 + 18 + 20}{4}$$

18.

$$z = \frac{82 - 70}{8}$$

19.

$$P(A) = \frac{3}{10}$$

20.

$$\mu = \frac{12 + 15 + 18 + 20}{4}$$

**B. Apply the key formulas**

1.

$$P(A) = \frac{\text{favorable}}{\text{total}}$$

2.

$$\mu = \frac{\sum b}{n}$$

3.

$$Z = \frac{c - \mu}{\sigma}$$

4.

$$P(A) = \frac{\text{favorable}}{\text{total}}$$

5.

$$\mu = \frac{\sum e}{n}$$

6.

$$Z = \frac{a - \mu}{\sigma}$$

7.

$$P(A) = \frac{\text{favorable}}{\text{total}}$$

8.

$$\mu = \frac{\sum c}{n}$$

9.

$$Z = \frac{d - \mu}{\sigma}$$

10.

$$P(A) = \frac{\text{favorable}}{\text{total}}$$

11.

$$\mu = \frac{\sum a}{n}$$

12.

$$Z = \frac{b - \mu}{\sigma}$$

13.

$$P(A) = \frac{\text{favorable}}{\text{total}}$$

14.

$$\mu = \frac{\sum d}{n}$$

15.

$$Z = \frac{e - \mu}{\sigma}$$

16.

$$P(A) = \frac{\text{favorable}}{\text{total}}$$

17.

$$\mu = \frac{\sum b}{n}$$

18.

$$Z = \frac{c - \mu}{\sigma}$$

19.

$$P(A) = \frac{\text{favorable}}{\text{total}}$$

20.

$$\mu = \frac{\sum e}{n}$$